Data Warehouses

Lecture – general Information

Contact details

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- Contact:
  - E-mail:
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      - Formal matters, requests and absences
  - MS Teams
    - Available during consultation hours
      - General matters, questions and comments
- Consultation hours:
  - Available in USOS

Meeting and resources

Additional software:

MS Teams

- chat, appointments

EPortal

- assignments, solutions and attendance
- grades

Course resources:

https://eportal.pwr.edu.pl/

Using AI Tools

- IT Department recommendations
  - Copyright Check:
    - Verify whether the content does not infringe on copyrights.
  - Labeling Requirement:
    - AI-generated content must be clearly marked/labeled.
  - Verification Requirement:
    - Necessity to verify the accuracy and correctness of the content.
  - Data Security Recommendation:
    - Do not provide personal data, confidential information, or data that must be protected (intellectual property).
  - Supportive Role:
    - AI tools should only serve as support during content creation.
  - Accountability:
    - The author bears full responsibility for the presented content, even if it was generated by AI.

Using AI Tools

- W4 Guidelines for Students
- Documentation of Use:
  - The author of a work (project documentation, diploma thesis, term paper, or any other work required for course credit) must describe the AI tools used and the purposes for which they were used (e.g., language editing, generating tables, generating a bibliography).
  - Instructor Approval:
    - The purposes and scope of using AI tools should be agreed upon with the course instructor.
  - Transparency of Process:
    - The author should allow the course instructor to view the prompts and AI tools responses used during the creation of the documentation.
  - Clear Labeling:
    - The author must explicitly mark fragments of the work (e.g., images, tables) created using AI tools within the documentation (e.g., in footnotes).
  - Student Responsibility:
    - The use of AI tools does not exempt the student from responsibility for:
    - Any type of errors.
    - Citing non-existent literary sources.
    - Unacceptable moral and ethical content.
    - Infringement of copyrights.

Using AI Tools

- Laboratory guidelines
  - The scope of AI tool usage must be precisely defined
    - The extent to which AI tools are used must be clearly and accurately specified.
  - AI tool usage must be conscious and intentional
    - A plan for applying AI to a given task must be developed, including the definition of evaluation criteria.
  - Evaluate application potential
    - An effort should be made to draw conclusions regarding the capabilities and the specific mode of application for a given tool.
  - AI tool usage must be critical
    - A critical analysis of the obtained results must be conducted.
  - Documentation and justification
    - The process of verifying the AI-generated responses must be documented and justified.

Using AI Tools

- University level:
  - https://di.pwr.edu.pl/oprogramowanie/ai/zalecenia-korzystania-z-narzedzi-ai
- Department level:
  - https://wit.pwr.edu.pl/en/students/guidelines-for-the-use-of-ai-tools

Course

Lab:

- 30h of class time
  - on average 2h a week of class
- 60h of total time
  - on average 4h a week

Lecture

- 30h of class time
  - on average 2h a week of class
- 60h of total time
  - on average 4h a week

In total 120h for the course

- 4h a week during class
- 4h a week outside classes

PREREQUISITES

- Basic knowledge of database system
  - with a particular focus on the relational model
- At least basic knowledge of SQL query language

Aim

Aim:

get to know how to organize, store, process and visualise large volumes of data for intelligent decision support.

Perspective:

data warehouse system

Skills:

SQL for data analysis; design and implementation of a data model for data analysis; data pipelines for data movement, transformation and integration; architecture of a data warehouse system

SUBJECT OBJECTIVES

- Has basic knowledge and skills of using SQL for data analysis, including grouping operators, and SQL aggregation and grouping functions.
- Has basic knowledge and skills in the area of transaction-oriented processing (OLTP) and analytic oriented processing (OLAP).
- Has basic knowledge and skills of using data warehouses.
- Knows basics of MS PowerPivot, MS SQL Analysis Services, MS SQL Integration Services and MS SQL Reporting Services.
- Has basic knowledge and skills in data integration, reporting and visualisation.

SUBJECT EDUCATIONAL EFFECTS

relating to knowledge:

- has basic knowledge on data warehouse usage and data warehouse organisation – logical and physical
- has basic knowledge on ETL process, reporting and data analysis

relating to skills:

- can use SQL grouping operators and SQL grouping and aggregating functions
- can design and implement a ETL process
- can design and implement a simple data warehouse and use it to generate basic reports, using different data visualisation methods
- can use basic MDX queries

Lecture

PROGRAMME CONTENT		
Form of classes - lecture		Number of hours
Lec 1	Course details. Introduction to Business Intelligence.	2
Lec 2	SQL grouping operators. SQL aggregating and grouping functions.	2
Lec 3	Transaction vs analytic needs, processes and data sources	2
Lec 4	Multidimensional data model – logical organization	2
Lec 5	Data warehouses – basics	2
Lec 6	ETL proces	2
Lec 7	Data warehouse – logical organisation	2
Lec 8	Data warehouses – architecture	2
Lec 9	MDX queries	2
Lec 10	MDX queries	2
Lec 11	Multidimensional data model – physical organisation	2
Lec 12	Reporting	2
Lec 13	Data visualisation	2
Lec 14	Data warehouse – design basics	2
Lec 15	Web dashboards	2
	Total hours	30

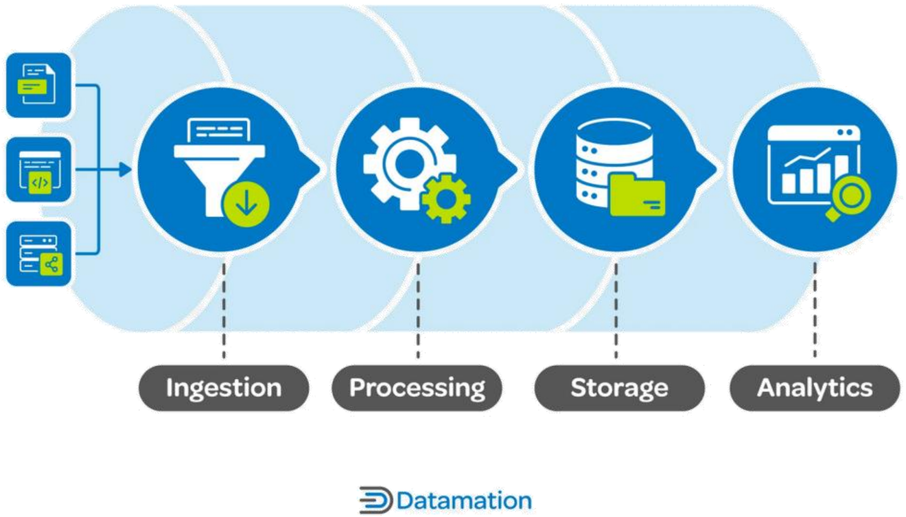
Lab

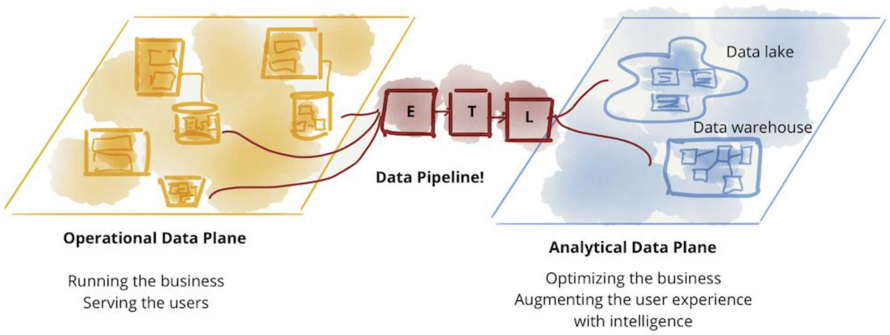
AIM:

- Implementation of DW in MS SQL Server infrastructure
  - Using AdventureWorks sample database
    - (available online)
- Basics of ETL processing using SQL Server Integration Services
- Cube implementation using SQL Server Analysis Services
- Using cubes in OLAP analysis
- Basics of data visualization and reporting

Mini-project Idea

- Prepare a complete technical implementation of an analytical data platform that follows a data warehouse architecture.
  - In particular, this platform should be able to tackle all of the identified OLAP needs of its users (through a set of reports, visualisations and dashboards) using the available data (originating from external sources/systems; available in different formats).
- An underlying assumption is that an intermediate storage solution, located between data sources and users, is utilised.
  - This storage is loaded with data, originating from external source systems, through a dedicated ETL solution.
  - ETL process involves staging extracted data in a separate relational database system.
  - Further the analytical storage is designed using the multidimensional model and is further limited to a single data mart solution, covering only a selected single line of business.
  - Implementation of the data mart requires a creation of a single dimensional data cube (using SSAS).
- Finally, user requirements are fulfilled through a set of well-designed and implemented reports, visualisations and interactive dashboards.





Mini-project Idea

GENERAL OVERVIEW

GENERAL OVERVIEW OF 1ST MINI PROJECT PART:

- Focus is on Analysis
- Find dataset and get yourself familiar with the data, perform basic data profiling
- Describe analytical needs against selected dataset, determine dimensions, fact, and measures

GENERAL OVERVIEW OF 2ND MINI PROJECT PART:

- Focus is on Design
- Design multidimensional model for the selected dataset
- Create Logical Data Map
- Plan and design the ETL process: extract, clean, transform, and load your data

GENERAL OVERVIEW OF 3RD MINI PROJECT PART:

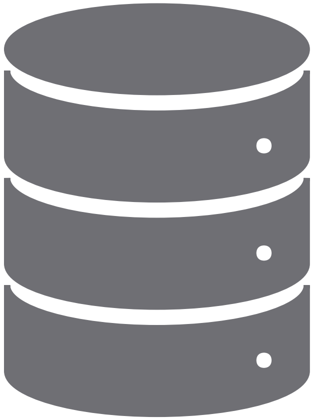
- Focus is on Implementation
- Implement the ETL process using SQL Server Integration Services SSIS
- Implement a multidimensional cube using SQL Server Analysis Services SSAS

GENERAL OVERVIEW OF 4TH MINI PROJECT PART:

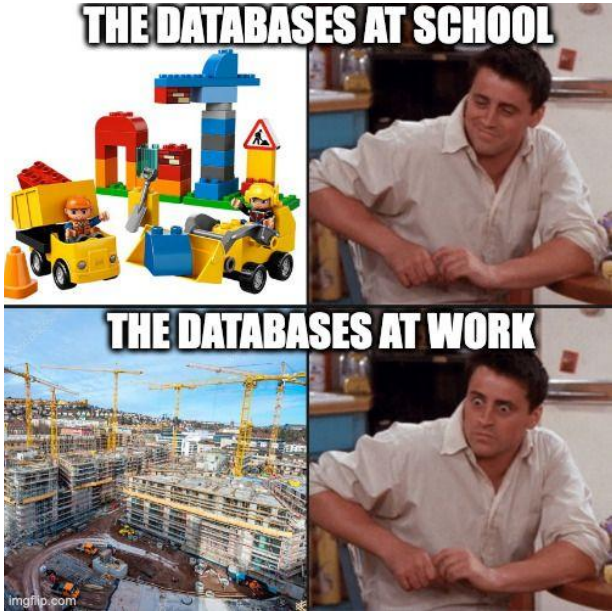
- Focus is on Usage
- Using the implemented cube perform basic data analysis and visualisation using Tableau/Power BI Desktop

Mini-project Idea

Mini-project Data

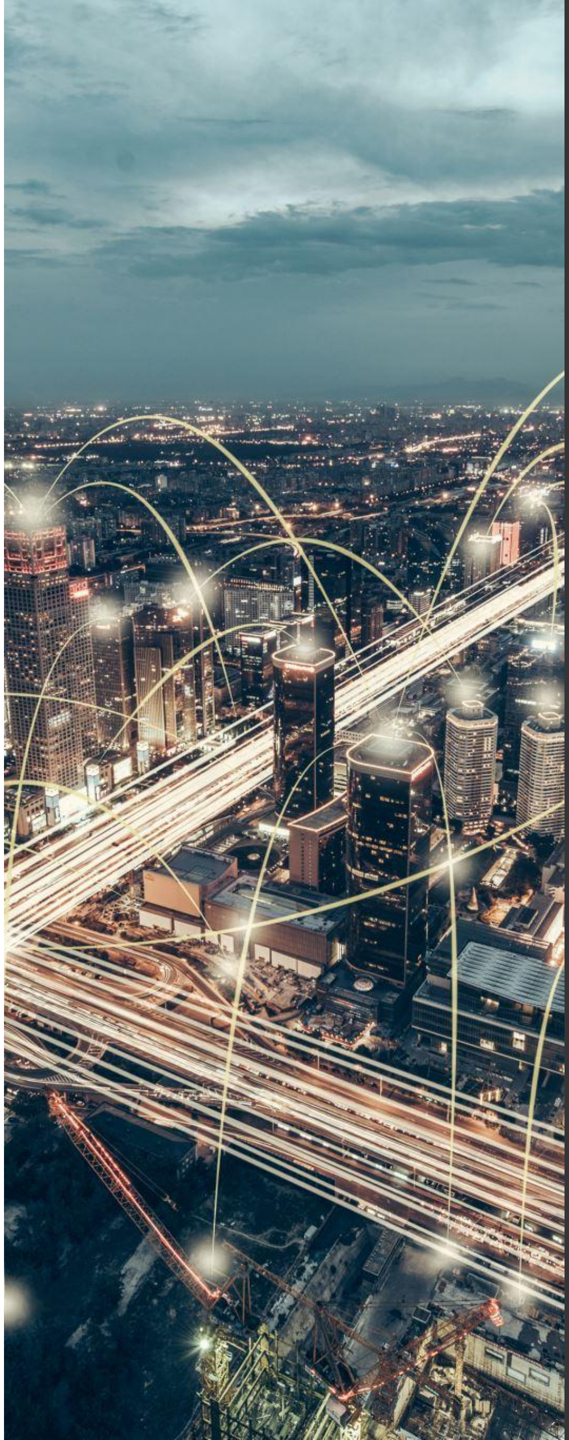


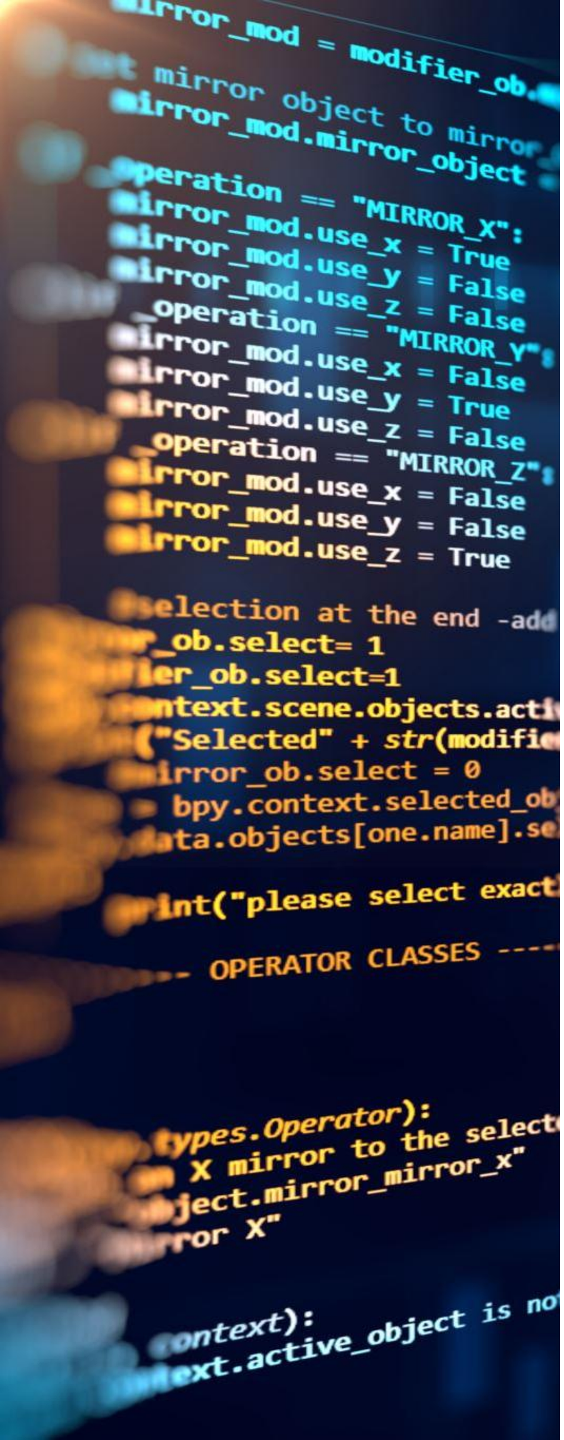
- Remember to find/select a database/data source(s) for your mini-project.
    - The data should allow for multidimensional analysis (OLAP)
    - As such, the data should be operational in nature and as close to data generation as possible
        - Try keeping the data close to its source
    - It would be good if you could see the data as a representation of a series of measurements (preferably quite a long one)
        - events (understood in a quite generic way), that can be measured (using certain numerical values, eg. sales) and analysed from multiple perspectives (dimensions, eg. client, product, etc.). It
- Examples
    - of events include: product sales, room rental, social net. comments, news posting, product returns, meeting attendance, item review, etc.
    - of measures include: sales amount, rental rate, comment's sentiment score, ad. revenue, processing time, late time, review score, etc.
    - of perspectives include: Product, Room, User, News outlet, Employee, Meeting, Item, etc.



Mini-project Data

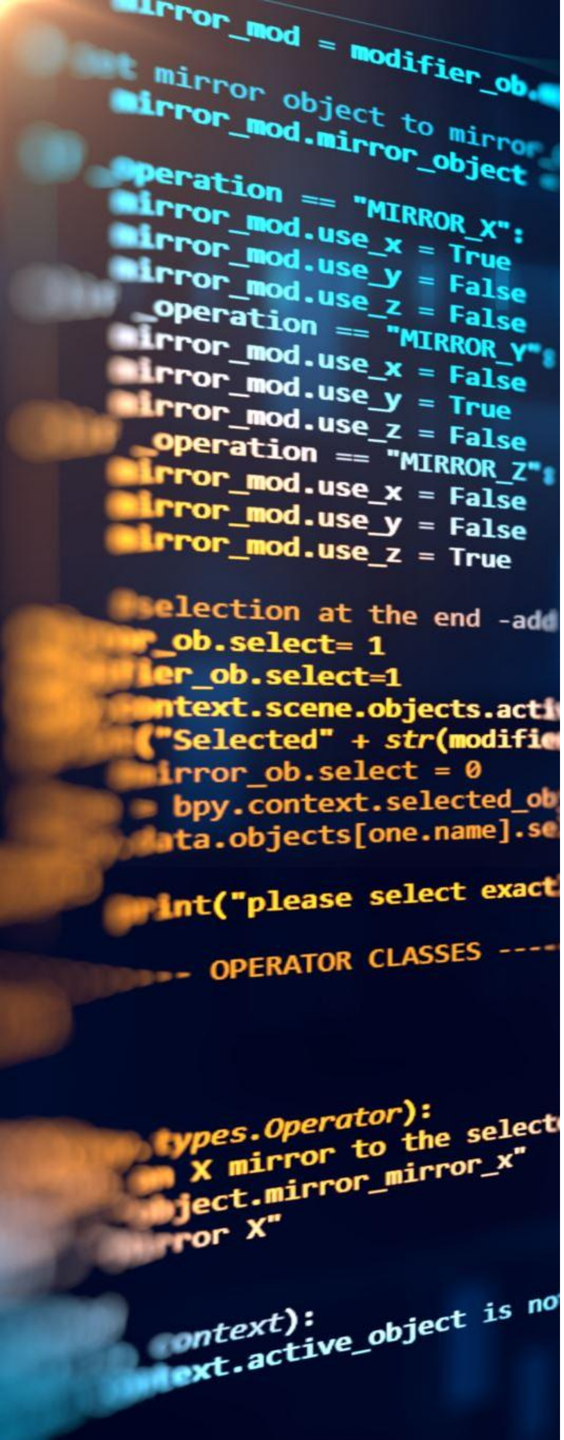
- Datasets characteristics to look for:
  - Look close, Look at domains you are interested in
    - Domain you can easily understand and play with
  - Originating from – at least – two data sources
    - e.g. NYC taxi rides and NYC bike share, sales data and weather data.
  - Operational data
    - Stay away from highlevel aggregates or highly processed analysis-ready data
  - Data should have enough attributes - should be rich-enough
  - Data can be a series of flat files (csv, tsv, txt), data dumps, database (any database), or other





Software

- Required software:
  - MS Windows 10/11
  - MS SQL Server 2025
    - Developer Enterprise Edition with:
      - Database Engine
      - Analysis Services
  - MS SQL Server Management Studio
  - Power BI Desktop
  - MS Visual Studio
    - With MS SQL Data Tools
      - requires Integration and Analysis Services Projects extensions
  - VS Code



Software

- Required sample databases:
  - AdventureWorks 2014
    - Two versions OLTP and DW
      - https://github.com/Microsoft/sql-server-samples/releases/tag/adventureworks

Grade

Lecture Grade

Exam - two terms

Multiple choice test with open questions

Lab Grade

Details during laboratory classes

- Final test
  - A mixture of multiple-choice test and a few open-ended questions
    - Multiple-choice with partial points, multiple answers can be correct
    - Open-ended questions are short descriptions
  - Scope:
    - Topics discussed during the lecture
    - Topics recommended for independent study
  - 4 parts – roughly 9 questions per part:
    - using SQL for data analysis
    - transaction-oriented processing (OLTP) and analytic oriented processing (OLAP).
    - data warehouses architecture and data models.
    - data integration, reporting and visualisation.
  - Passing the test:
    - Obtaining a minimum of 50% of the points from <u>each</u> part

- Example
  - Test question:
    - What makes analysis of operational data difficult? (Mark 1 for true, 0 for false)
      - A High denormalization of source data structures
      - B Inconsistent definitions of data in data sources
      - C Different modelling requirements for source data
      - D High stress on source data systems
  - Open-ended question:
    - **Comment on the topic:** Data in data warehouse directly represents operational data and processes

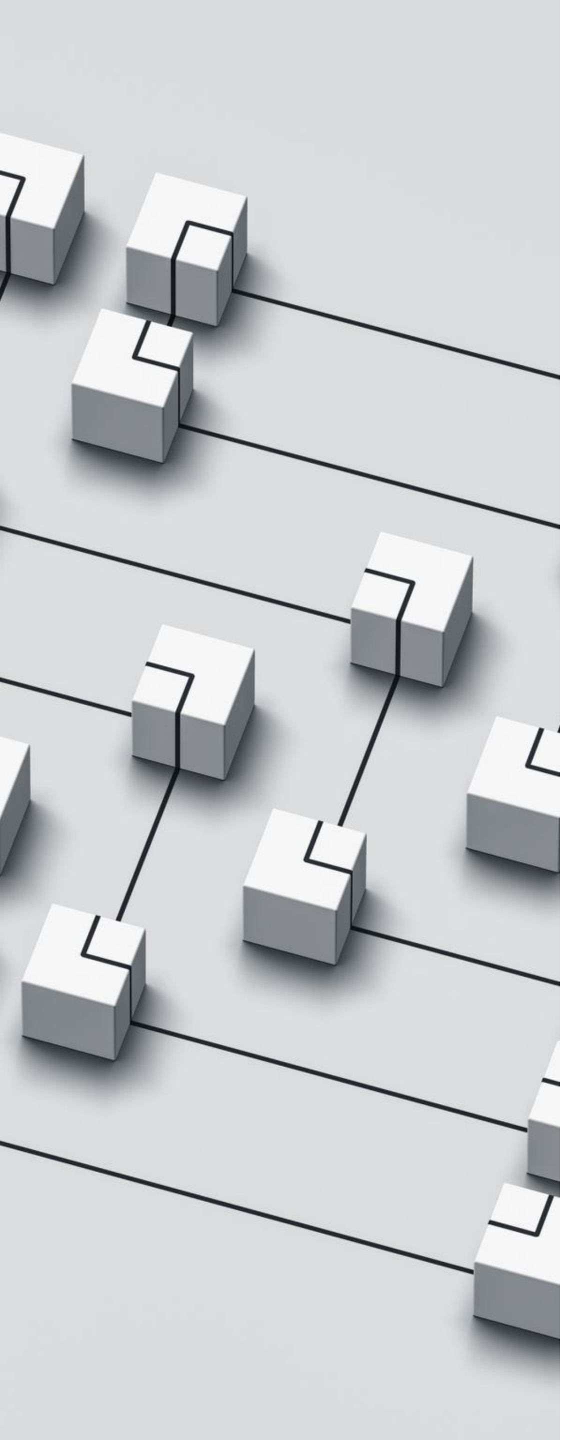
Final grade

To pass

- Lecture
  - at least 50% from each part of the test
  - then the total % is your final score

Final grade

- 50-59% – 3.0
- 60-69% – 3.5
- 70-79% – 4.0
- 80-89% – 4.5
- 90-100% – 5.0



Literature

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Formal literature

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